

MAN403 - SELECTED TOPICS IN MANAGEMENT SCIENCE

Course Code - Name	: MAN403 - Selected Topics in Management Science (elective)
Theoretical course hours	: 3 per week
Local credit	: 3
ECTS	: 5 (by EU standards, equals $5 * '25-30' = '125-150$ hours' of effort) (by H.U. standards, equals $5 * 30 = 150$ hours of effort)
Prerequisites:	: MAN305 Operations Research II is assumed to be known: linear and mixed-integer programming, modelling with Gurobi in Python. Dynamic programming is also assumed to be known. The course is taught and all projects are coded in Python.
Contact	: <u>Course Material</u> : https://hadi.hacettepe.edu.tr (Moodle system) <u>Announcements</u> : http://www.isletme.hacettepe.edu.tr <u>Office Hours</u> : See the schedule announced by the instructor for the office hours, when you can freely communicate directly to the instructor.
Course objective	: To understand and use advanced operations research techniques for complex organisational problems, and to gain experience in applying them with software. When a problem is too large or too uncertain to be solved exactly, a good heuristic is the practical answer; the course studies three families: heuristics built on LP/MIP, on dynamic programming, and on simulation and machine learning.
Learning outcomes	: Following outcomes are expected: • Students can design LP/MIP-based heuristics (matheuristics), including decomposition and relaxation methods. • Students can design restricted and simulation-based dynamic programming heuristics. • Students can formulate a sequential decision problem as a Markov decision process and solve it with temporal-difference (TD) learning. • Students can evaluate and compare heuristics through simulation experiments and report the results in an academic style.
Material to read	: Textbooks: Sutton, R. S. & Barto, A. G. (2018) "Reinforcement Learning: An Introduction" (2nd ed.), Cambridge, MA: MIT Press (freely available from the authors). Sidebooks: Winston, W. L. (2004) "Operations Research: Applications and Algorithms", Belmont, CA: Thomson Brooks/Cole. Powell, W. B. (2011) "Approximate Dynamic Programming: Solving the Curses of Dimensionality" (2nd ed.), Hoboken, NJ: Wiley. Maniezzo, V., Stützle, T. & Voß, S. (Eds.) (2010) "Matheuristics: Hybridizing Metaheuristics and Mathematical Programming", New York: Springer. Software: Python; Gurobi Optimizer with a free academic license and <code>gurobipy</code> ; Visual Studio Code as the recommended editor. Articles on each project problem are listed in the project booklet.
Exercising	: Reproduce the in-class algorithms in Python and apply each one to your own project problem as the weeks go on.

Student Workload and ECTS Calculation

	Activities	Number	Duration (hours)	Total Work Load (hours)
Time Management	Project	2	20	40
	Preparation for the Midterm Exam	1	10	10
	Course Attendance	14	5	70
	Preparation for the Final Exam	1	40	40
	Total			160
Evaluation	Project 1 (LP/MIP-based heuristic)		15%	
	Project 2 (DP-based heuristic)		15%	
	Project 3 (simulation / TD-learning heuristic)		15%	
	Attendance		5%	
	Final (comparative report and oral exam)		50%	
	Explained in detail below.			

Grading Policy:

- Your final grade will be the weighted sum of three projects ($3 \times 15\%$), attendance (5%) and the final (50%). There are no bonuses.
- **One company per student.** In the first week every student is assigned one of the eight companies in the project booklet, and keeps it for the whole semester. Each company faces a decision problem that repeats over time and involves uncertainty (inventory, routing, inventory-routing and scheduling). The booklet fixes the size of each problem; you choose the numbers (demands, costs, distances, ...) yourself and explain your choices. Plain Python (lists, loops, **if** statements) is enough for the projects.
- **Projects.** The course has three blocks, each on one family of solution techniques. At the end of each block you submit a project that solves *your* problem with that block's technique:
 - Project 1: an LP/MIP-based heuristic (e.g. rolling horizon, fix-and-optimize, decomposition, relaxation), **due in Week 6**.
 - Project 2: a restricted or simulation-based dynamic programming heuristic, **due in Week 10**.
 - Project 3: a temporal-difference learning heuristic, **due in Week 14**.

Each project consists of your Python code and a short report in the style of an academic paper (the company and your numbers, your method, your results, references). Every method is measured by running your simulator **100 times** on the same 100 random seeds, as described in the booklet, so that your own three methods are compared on equal terms.

- **Project Market.** In Week 14 a “Project Market” session is held: each student presents a poster on their company and the three heuristics, to an audience of invited department students and alumni.
- **Final (50%).** The final has two parts:
 - a **comparative report**: you compare the heuristics you wrote in the three projects. Since every student has a different company and different numbers, costs are not comparable across students — what is comparable is *how much each method improved on its own simple rule* and *what it cost in computing time*. Using those two measures, and your classmates' reports (shared with the class after each deadline), you discuss which kind of heuristic works for which kind of problem, and why;
 - an **oral exam**: you present your report and answer the instructor's questions on the heuristics.

The dates of the report submission and the oral exams will be announced by the department.

- Explain your work **in your own words**. It is allowed to use online AI services, yet it is allowed to use them, not to obey them. Any material you use from others (code, figures, text, ideas) must be cited.

Class Policy:

- Students are obliged to maintain the learning environment.
- When asked, students are expected to participate in in-class discussions and activities.
- Lack of knowledge of the rules is not a reasonable explanation for a violation. Any violation of the rules or causing a disturbance in the classroom will be observed and necessary actions will be taken.

Tentative Course Outline:

- ▶ **Lecture -1-** Introduction: heuristics for hard, uncertain decision problems
-22.09.26-
 - * Introducing the lecturer, course process and rules.
 - * Why exact methods fail: size and uncertainty; policies instead of plans.
 - * The eight project companies; assignment of companies to students.
 - * Testing a decision rule by simulation: 100 runs, the same seeds for every method.

Block 1: LP/MIP-based heuristics

- ▶ **Lecture -2-** Rolling Horizon and Fix-and-Optimize
-29.09.26-
 - * Deterministic (expected-value) models as a basis for stochastic problems.
 - * Rolling-horizon planning; relax-and-fix and fix-and-optimize.
- ▶ **Lecture -3-** Splitting a Big Decision into Smaller Ones
-06.10.26-
 - * Splitting one big decision into smaller ones that can be solved in turn.
 - * Decomposing by time (period by period), by geography (cluster first, route second), and by decision type (assign first, sequence second).
 - * Feeding the pieces back into each other; where decomposition loses quality.
- ▶ **Lecture -4-** Dropping the Hard Constraint, then Repairing
-13.10.26-
 - * Dropping a hard constraint to get an easy model and a bound; repairing the solution afterwards.
 - * Integrality relaxation and rounding; turning hard constraints into penalties.
 - * Using bounds to know how far a heuristic can still be improved.
- ▶ **Lecture -5-** Matheuristics and ALNS; Project Workshop 1
-20.10.26-
 - * Adaptive large neighbourhood search with MIP-based repair.
 - * Workshop: Project 1 progress.

Block 2: DP-based heuristics

- ▶ **Lecture -6-** Restricted Dynamic Programming
-27.10.26-
 - * The curse of dimensionality; keeping only the most promising states at each stage.
 - ⇒ *Project 1 due.*
- ▶ **Lecture -7-** Restricted Dynamic Programming: Design Choices
-03.11.26-
 - * State definitions, selection (restriction) rules, dominance, the restriction width.
- ▶ **Lecture -8-** Simulation-based Restricted Dynamic Programming
-10.11.26-
 - * Estimating the value of a state under uncertainty by sampled scenarios.
 - * Sample size, common random numbers, and the cost of simulation.
- ▶ **Lecture -9-** DP-based Heuristics: Applications; Project Workshop 2
-17.11.26-
 - * Rollout and look-ahead policies as relatives of restricted DP.
 - * Workshop: Project 2 progress.

Block 3: Simulation- and learning-based heuristics

- ▶ **Lecture -10-** From MDPs to Learning: Monte Carlo and Temporal-Difference Methods
-24.11.26-
 - * Learning a value function from simulated experience; Monte Carlo vs TD(0).
 - ⇒ *Project 2 due.*
- ▶ **Lecture -11-** TD Control: SARSA and Q-learning
-01.12.26-
 - * On-policy and off-policy TD control; exploration.
- ▶ **Lecture -12-** TD Learning with Function Approximation
-08.12.26-
 - * Large state spaces: features and linear value functions; semi-gradient TD.
- ▶ **Lecture -13-** TD Learning: Applications; Project Workshop 3
-15.12.26-
 - * Training, tuning and testing a learned policy.
 - * Workshop: Project 3 progress; poster preparation.
- ▶ **Week -14-** *Project Market*
-22.12.26-
 - * Poster presentations of the problems and the three heuristics.
 - ⇒ *Project 3 due.*